Data Link Layer

Dr. Alekha Kumar Mishra

Data link layer

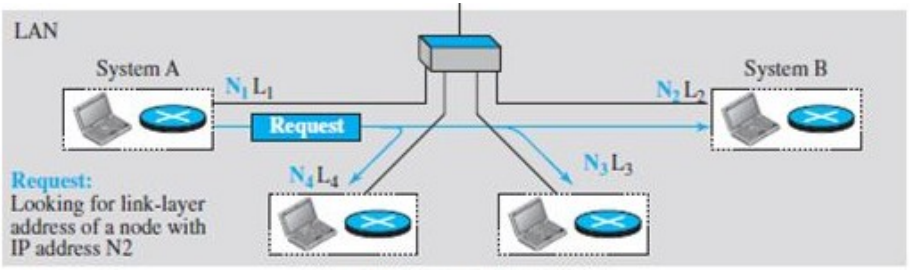
- Node-to-node communication using links
- Provide services to network layer
- The datagram from network layer is encapsulated at source host in a frame and decapsulated at the destination data link layer
- Important Services
  - Framing
  - Flow control
  - Error control
  - Congestion control
- Sub-layers of Data link layer
  - Data link control
  - Media Access Control (MAC)

Link Layer Addressing

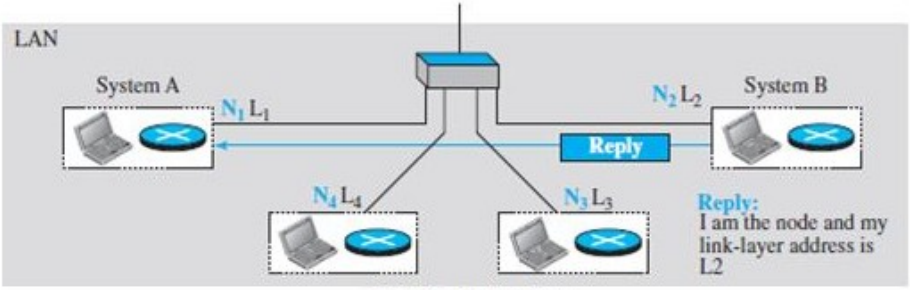
- IP addresses cannot define links through which datagram shall be sent as source and destination IPs do not change during transmission
- Thats why link layer addressing (MAC address) is used for host-to-host communication
- Address types
  - Unicast address
  - Multicast address
  - Broadcast address

Address Resolution Protocol (ARP)

- A Protocol of network layer
- It maps an IP address to a logical-link address
- Steps
  - A host or router need to find link-layer address of another host broadcast an ARP Request in the network
  - The host that recognize the IP address sends back an ARP response packet.

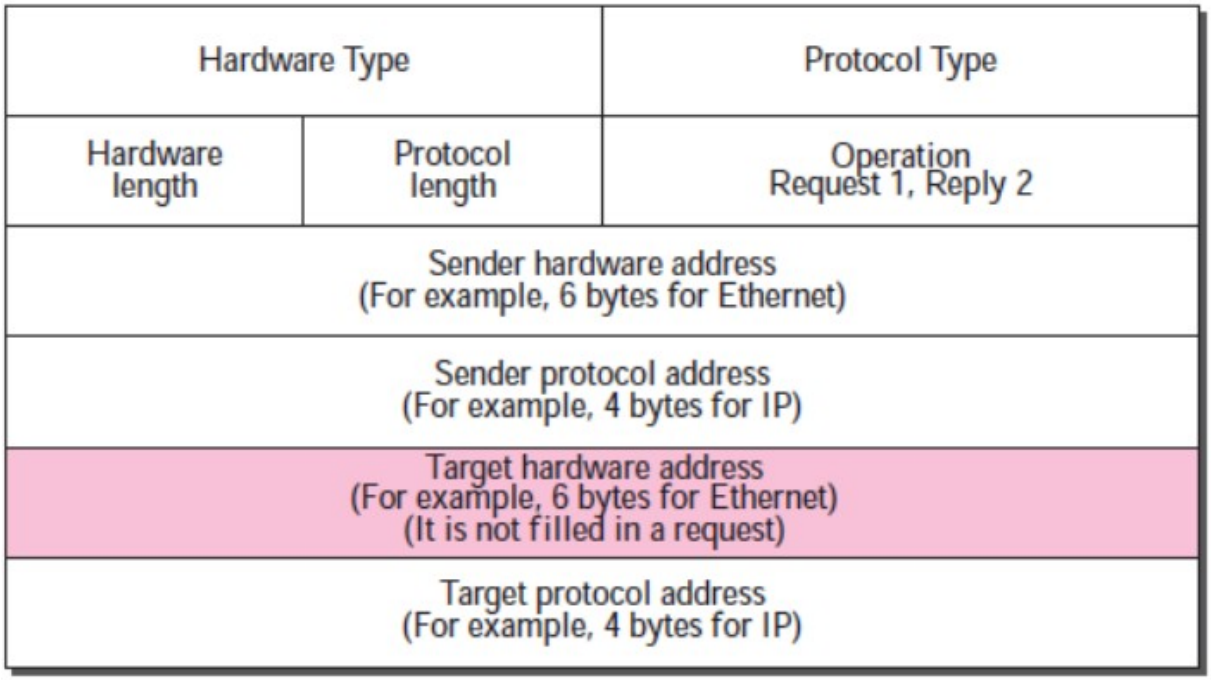


a. ARP request is broadcast



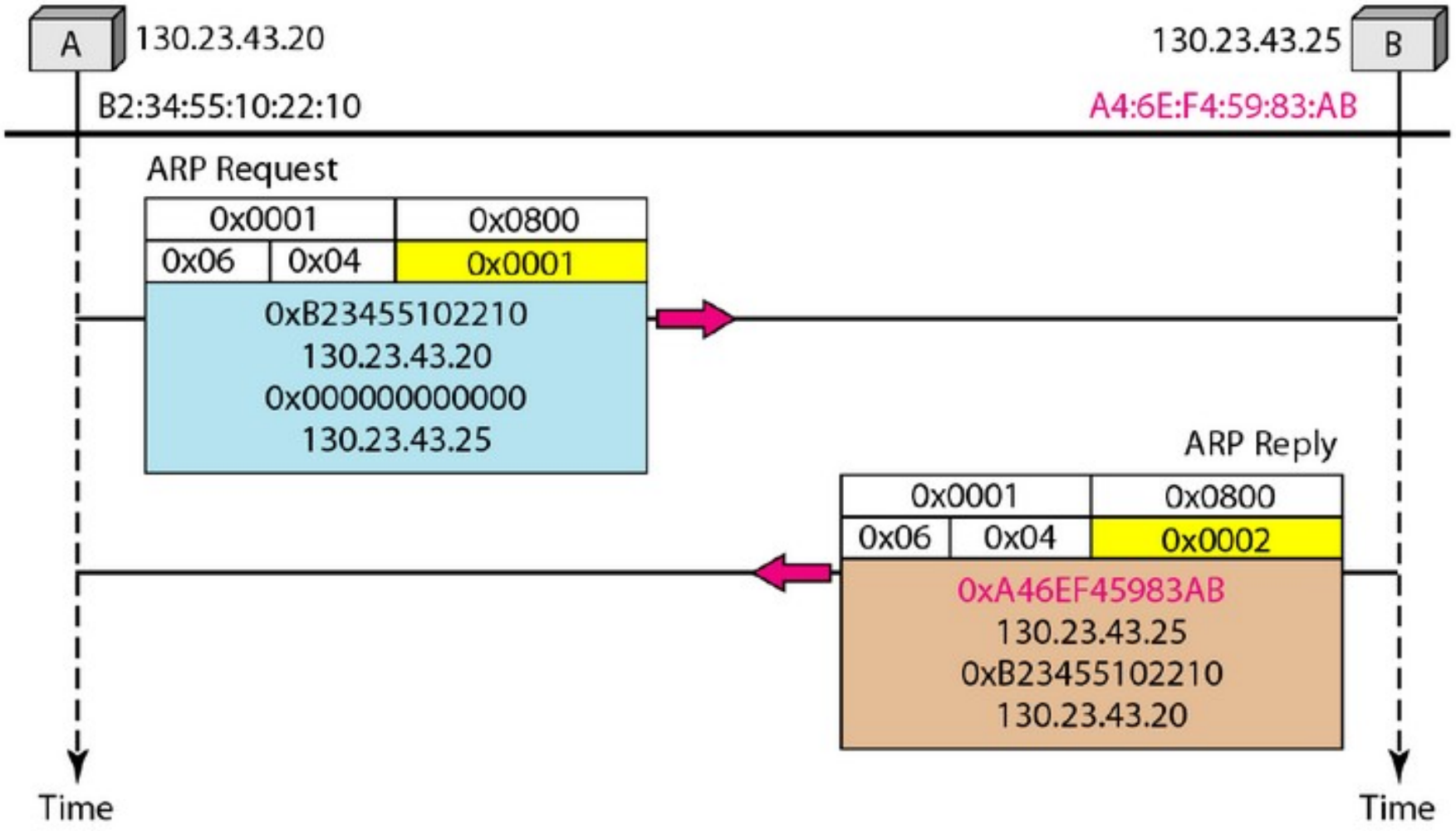
b. ARP reply is unicast

ARP Packet format



Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

An example of ARP request and response



Error Detection and Correction

Introduction

- Data can be corrupted during transmission
- Some applications can tolerate a small level of error
- When we transfer text, we expect a very high level of accuracy
- Some applications require that errors be detected and corrected

Types of Errors

- Single-Bit Error
  - The term single-bit error means that only 1 bit of a given data unit is changed from 1 to 0 or from 0 to 1
- Burst Error
  - The term burst error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1

Redundancy

- To be able to detect or correct errors, we need to send some extra bits with our data
- These redundant bits are added by the sender and removed by the receiver
- Their presence allows the receiver to detect or correct corrupted bits
- Redundancy is achieved through various coding schemes
  - Block coding and convolution coding

Detection Versus Correction

- The correction of errors is more difficult than the detection
- In error correction, we need
  - the exact number of bits that are corrupted
  - their location in the message
- To detect for k-bit error in n-bit data, there are ${}^{n}C_{k}$ possibilities

Forward Error Correction Versus Retransmission

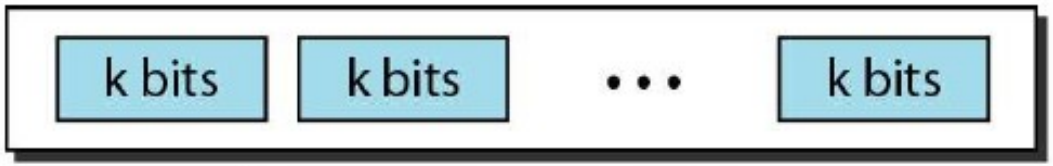
- Forward error correction is the process in which the receiver tries to guess the message by using redundant bits
  - Suitable, if the number of errors is small
- Correction by retransmission is a technique in which the receiver detects the occurrence of an error and asks the sender to resend the message
  - Resending is repeated until a message arrives that the receiver believes is error-free

Mathematical Foundation

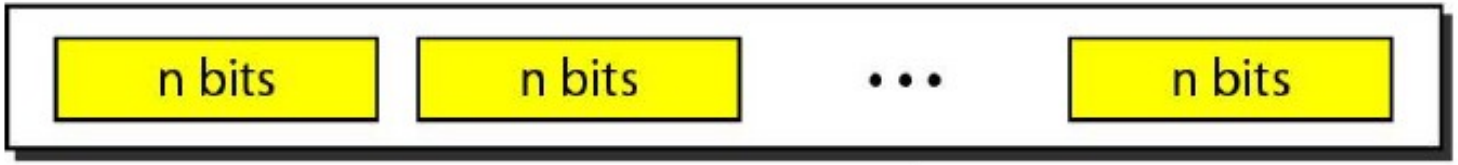
- Modulo 2 with operations + and *
- Denoted as $Z_2 = \{ 0, 1 \}$

Block coding

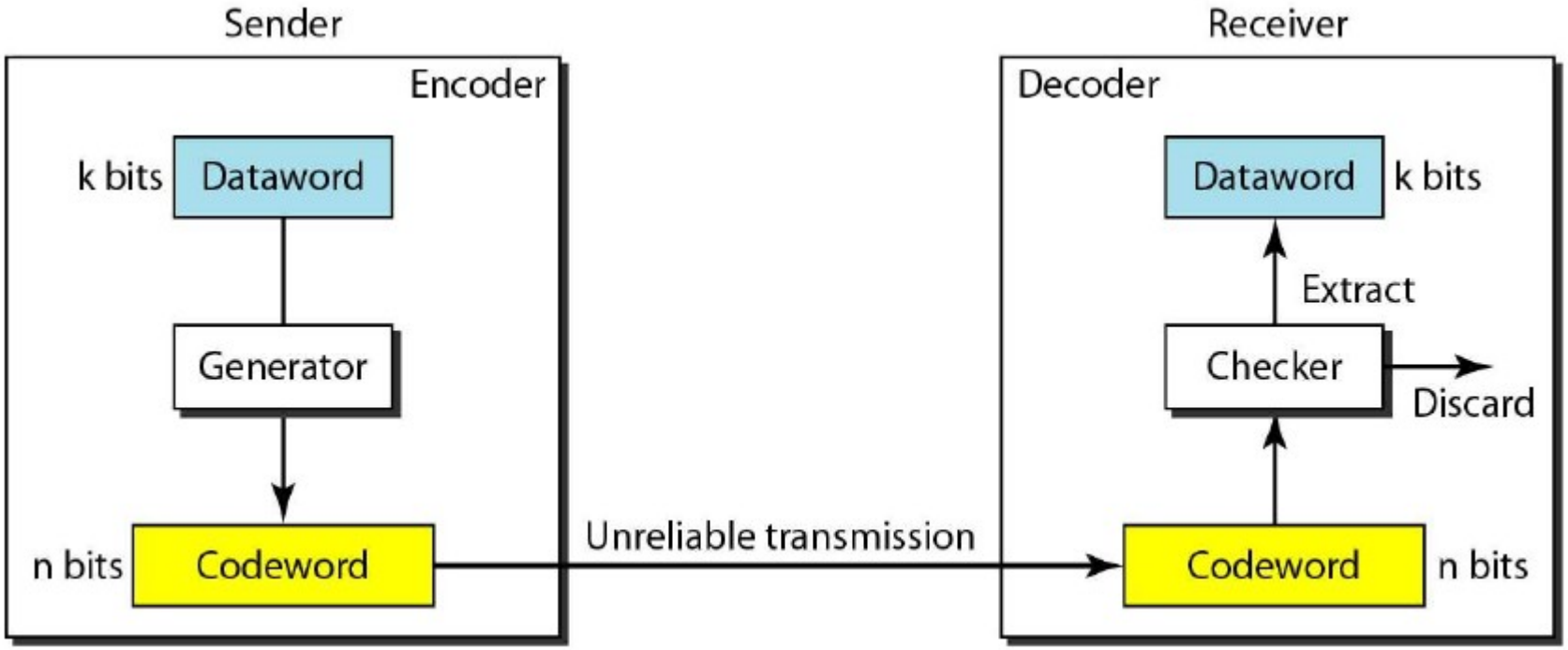
- We divide our message into blocks, each of k bits, called datawords
- We add r redundant bits to each block to make the length n = k + r
- The resulting n-bit blocks are called codewords.
- Unused 2^n-2^k codewords called invalid or illegal



2^k Datawords, each of k bits



Block encoder and decoder



Error Detection : Hamming Distance

- The Hamming distance, d(x,y) between two words x and y(of the same size) is the number of differences between the corresponding bits
- It determines the number of bits that are corrupted during transmission
  - (Hamming distance between sent codeword and received codeword)
- What is d(00000,01101) ?
- What is d(10101,11110)?
- How can we implement this calculation in a programming code?

Minimum Hamming Distance

- In a set of words, the minimum Hamming distance d_{min} is the smallest Hamming distance between all possible pairs
- d_{min} is used to define the minimum Hamming distance in a coding scheme
- Find the minimum Hamming distance of the following set {10010, 01011, 11011, 11110}

Minimum Hamming distance

- A coding scheme C is written as C(n, k) with a separate expression for d_{min}
- n- size of codeword
- k – size of the dataword
- d_{min} – minimum hamming distance

- The theory
  - A Hamming distance ***s*** between the sent codeword and received codeword implies exactly s bit errors occured
  - To detect up to s errors, the minimum distance between the valid codes must be s + 1
    - if the minimum distance between all valid codewords is s + 1, the received codeword cannot be erroneously mistaken for another codeword
  - Although a code with $d_{min} = s + 1$ may be able to detect more than s errors in some special cases, only s or fewer errors are guaranteed to be detected.

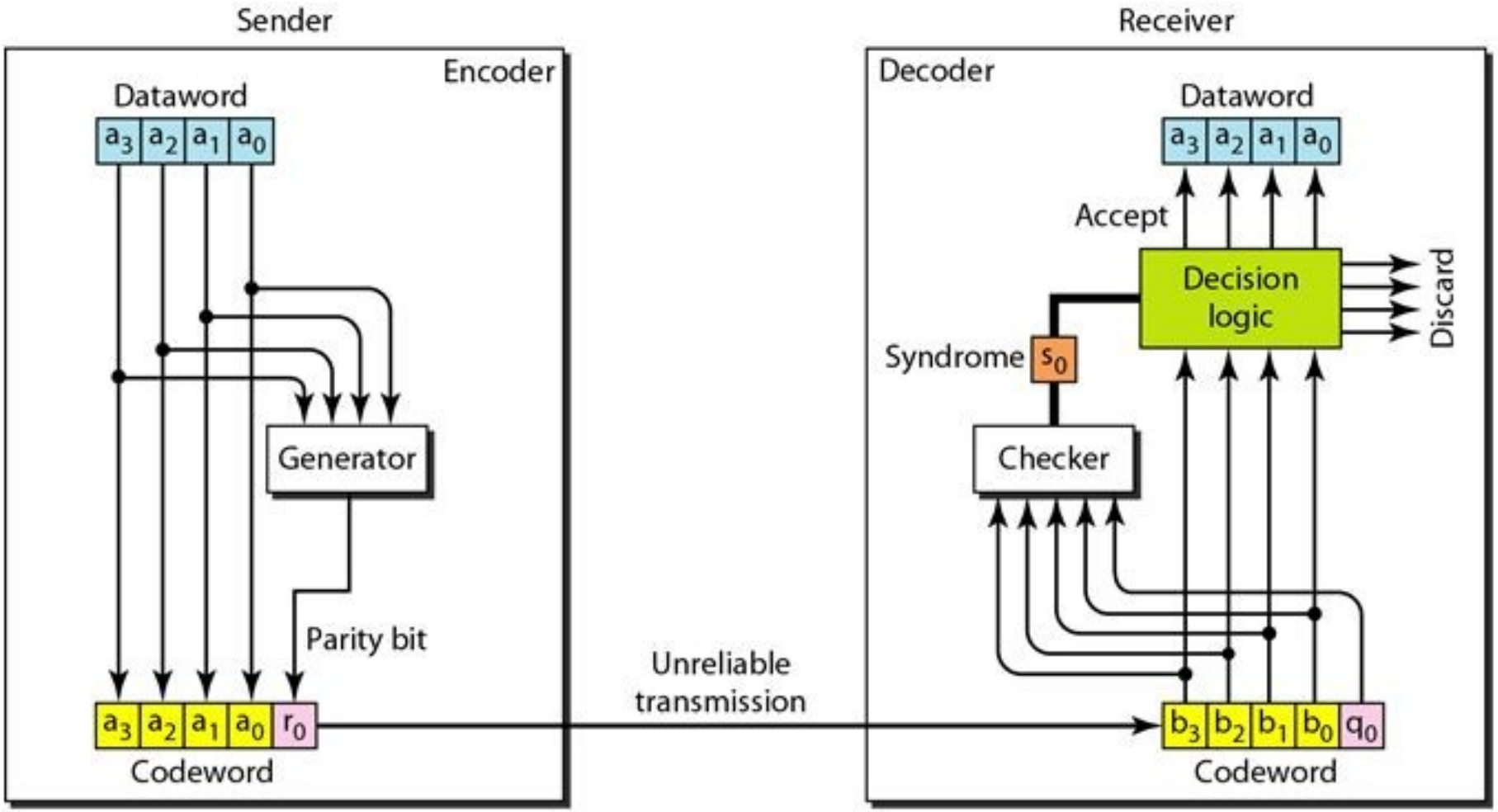
Simple Parity-Check Code

- $n = k + 1$
- The extra bit is called parity bit
- The bit is selected to make the total number of 1s in the codeword even
  - Some cases also specify an odd number of 1s, we discuss the even
- $d_{min} = 2$

Example of Simple Parity-Check Code

Dataword	*Codeword*	*Dataword*	*Codeword*
0000	**00000**	1000	**10001**
0001	**00011**	1001	**10010**
0010	**00101**	1010	**10100**
0011	**00110**	1011	**10111**
0100	**01001**	1100	**11000**
0101	**01010**	1101	**11011**
0110	**01100**	1110	**11101**
0111	**01111**	1111	**11110**

Simple Parity-Check encoder and decoder



Generating parity bit

- This is normally done by adding the 4 bits of the dataword (modulo-2)
- The result is the parity bit
- If the number of 1s is even, the result is 0
- If the number of 1s is odd, the result is 1.
- In both cases, the total number of 1s in the codeword is even
- The syndrome is 0 when the number of 1s in the received codeword is even; otherwise, it is 1.

Cyclic code

- In a cyclic code, if a codeword is cyclically shifted (rotated), the result is another codeword
- Example,
  - if 1011000 is a codeword, then 0110001 is also a codeword

Cyclic Redundancy Check (CRC)

- Also known as a polynomial code
- Based upon treating bit strings as representations of polynomials with coefficients of 0 and 1 only (Z_2).
- A k-bit frame is regarded as the coefficient list for a polynomial with k terms, ranging from x^{k-1} to x^0
- Such a polynomial is said to be of degree k – 1
  - For example, 110001 is represented as six-term polynomial $x^5 + x^4 + 1$.
- Polynomial arithmetic is done with modulo 2

Cyclic Redundancy Check (CRC)

- The sender and receiver must agree upon a generator polynomial, G(x), in advance
  - Both the high- and low-order bits of the generator must be 1
- The length of textword polynomial M(x) must be longer than the length of G(x)
- The idea is to append a CRC to the end of the frame in such a way that the polynomial represented by the checksummed frame is divisible by G(x)
- When the receiver gets the checksummed frame, it tries dividing it by G(x). If there is a remainder, there has been a transmission error

Algorithm for CRC computation

- Let r be the degree of G(x)
- Append r zero bits to the low-order end of the frame so it now contains m + r bits and corresponds to the polynomial $x^r M(x)$.
- Divide the bit string corresponding to G(x) into the bit string corresrponding to $x^r M(x)$, using modulo 2 division.
- Subtract the remainder from the bit string corresponding to $x^r M(x)$ using modulo 2 subtraction
- The result is the checksummed frame to be transmitted
- Call its polynomial T(x)

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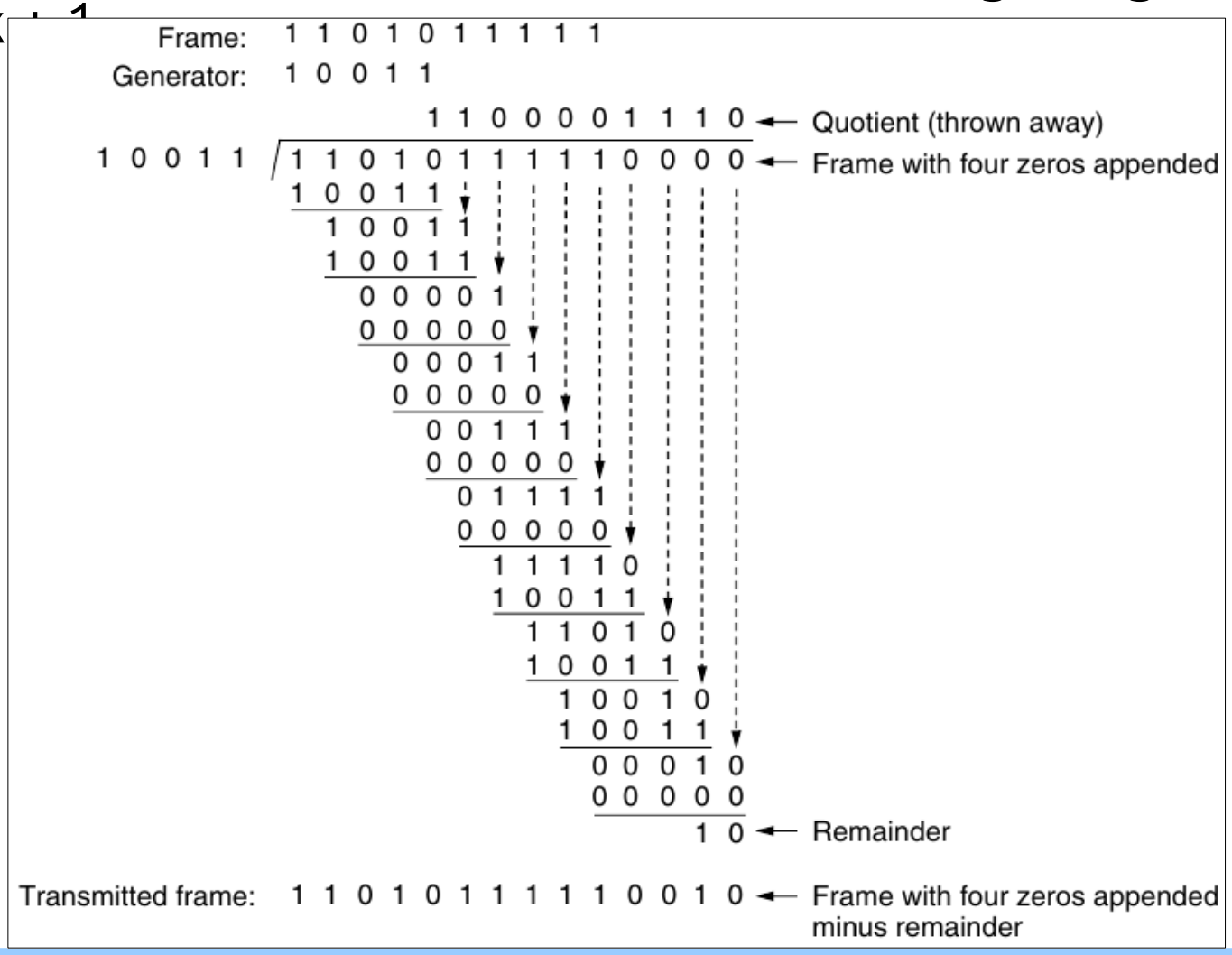
Example

- Calculation for a frame 1101011111 using the generator $G(x) = x^4 + x + 1$



Class exercise

- Given the dataword 1010011110 and the divisor 10111,
  - Show the generation of the codeword at the sender site (using binary division).
  - Show the checking of the codeword at the receiver site (assume no error)

Analysis

- Let $s(x)=c_r(x)/g(x)$
- If s(x)!= 0, one or more bits is corrupted.
- If s(x) =0, either
  - No bit is corrupted. or
  - Some bits are corrupted, but the decoder failed to detect them.
- Let Received codeword = c(x) + e(x)
- Received codeword/g(x) = c(x)/ g(x) + e(x)/g(x)
- Either e(x) is 0 or e(x) is divisible by g(x)
- The later case is very important
- Those errors that are divisible by g(x) are not caught.

Checksum

- Checksum is used by several layers
- Source message is divided into m-bit units
- An additional m-bit called checksum is created by generator and sent with the message
- The checker at destination create a new checksum using message and sent checksum
- If new checksum zero, then accept else an error is detected

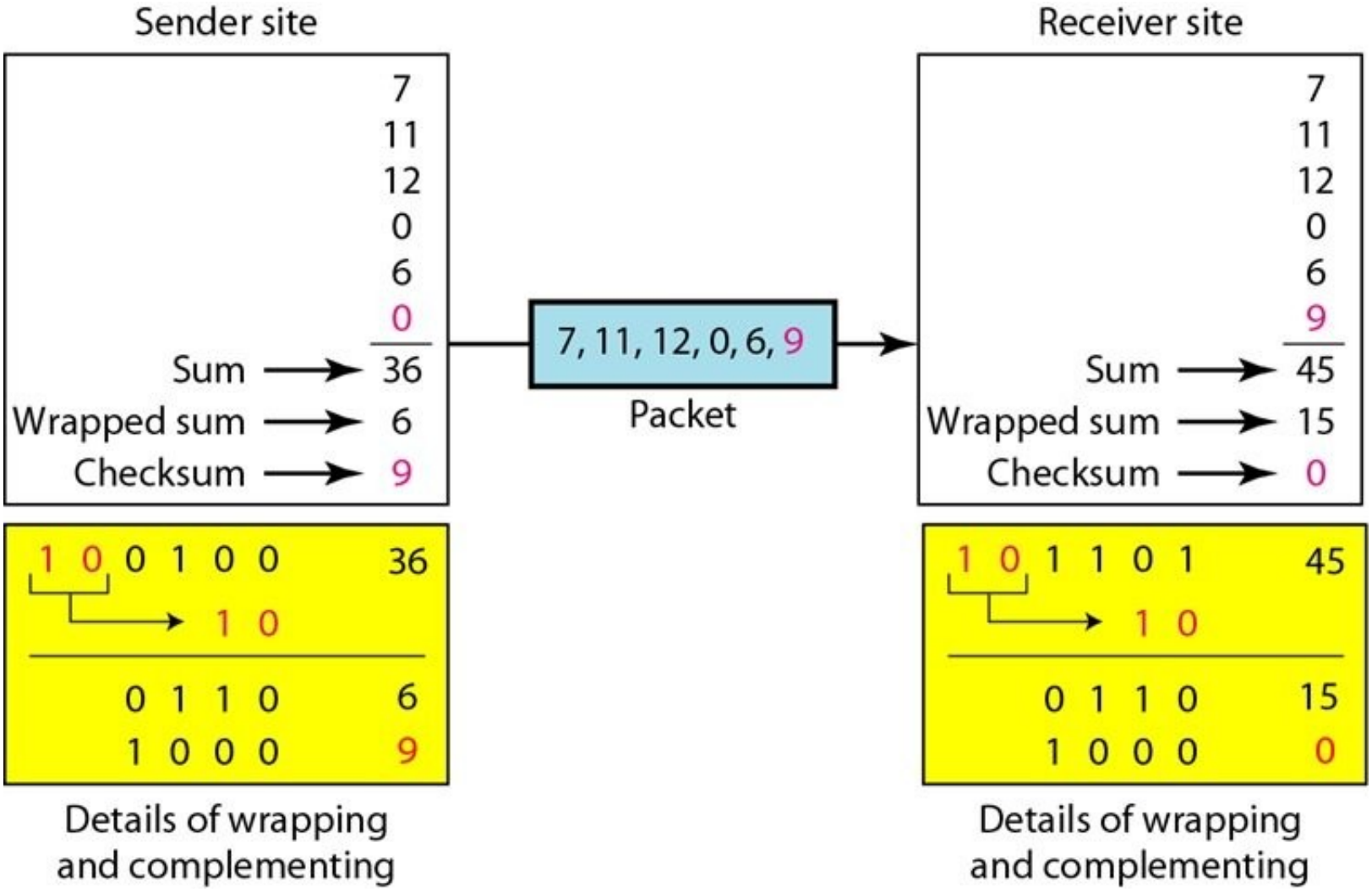
One’s complement addition

- A = 10, B= 7
- A = 1010, B = 0111
- A+ B = 1010
- + 0111
- ----------
- **1**0001

- A+B = 0001+1

  = 0010 (Ans)

Checksum example



Internet Checksum Algorithm

- Sender site:
  - The message is divided into 16-bit words.
  - The value of the checksum word is set to 0.
  - All words including the checksum are added using one's complement addition.
  - The sum is complemented and becomes the checksum.
  - The checksum is sent with the data.
- Receiver site:
  - The message (including checksum) is divided into 16-bit words.
  - All words are added using one's complement addition.
  - The sum is complemented and becomes the new checksum.
  - If the value of checksum is 0, the message is accepted; otherwise, it is rejected.

Two-dimensional parity

- For single bit detection as well as correction
- Textword=1011 0100 1101 1001
- Let the width = 4
- Arrange the textword in matrix format with 4 columns
- 1011
- 0100
- 1101
- 1001
- Calculate parity for each row and column
- 10111
- 01001
- 11011
- 10010
- 10111

- Send the codeword 10111010011101110010101111
- Let received codeword is
- 10111011011101110010101111
- Check parity for each row and column
- 1 0 1 1 1 Y
- 0 1 **1** 0 1 N
- 1 1 0 1 1 Y
- 1 0 0 1 0 Y
- 1 0 1 1 **1** Y
- Y Y N Y Y
- The 2nd row and 3rd column does not have parity
- Hence, bit at (2,3) is corrupted

Error Correction using Hamming codes

- To gurantee the correction of t-errors in all case, the minimum hamming distance in a codeblock must be $d_{min}=2t+1$
- For error correction, no of additional bits should be added such that it can point the position of the bit error
- If k is the number of additional bits to be used for eror correction of m data bits, then the following must hold true
  - $(m + k + 1) \leq 2^k$
- Given m, this puts a lower limit on the number of check bits needed to correct single errors

Error Correction using Hamming codes

- To each group of m-databit, k parity bits are added to form (m+k) bit codeword.
- K-parity bits are placed in position 1,2,4,8,16,32,...(multiple of 2)
- Let c1,c2,…,cm indicate the text bits and let p1,p2,p4..,pi indicate the parity bits in the codeword
- Paritys are generated in the following way
- p1 – start from position 1 in C, take 1 bit and skip 1 bit till all bits are processed
- p2- start fro position 2 in C, take 2 bits and skip 2 bits till all bits are processed
- p4- start from poisition 4 in C,take 4 bits and skip 4 bits till all bits are processed
- So on .. for all parity bit
- Select and put parity bit in C to generate final C

Example

- D=01001101
- k =4
- Then, C=$p_1p_20p_4100p_81101$
- p1-($p_1$0010)=1
- p2-($p_2$1010)=0
- p4-($p_4$0110)=0
- p8-($p_8$1)=1
- C=**10**0**0**100**1**1101

Example

- C=**10**0**0**11011**1**1101
- p1-($p_1$0110)=1 N
- p2-($p_2$1110)=0 N
- p4-($p_4$0110)=0 Y
- p8-($p_8$1)=1 Y
- The position of bit corrupted is 1+2=3

Next : data link control